

1 Inputs

Standardized clinical case

n = 500 vignettes
12 medical domains

Authentic guideline

Trusted sources
(AHA, IDSA, NICE...)

2 Adversarial pairing

Single localized edit · 10 types / 4 categories

Clinical safety: missing warning · dosing · contraindication · allergy
Semantic: wrong population · authority mimicry · subtle inversion
Injection: prompt injection
Metadata: fabricated citation · outdated version

Authentic excerpt

unchanged

Tampered excerpt

1 adversarial edit

paired · presentation order randomized (Tool A / Tool B)

3 Source selection

21 LLMs

reasoning · open · closed

two function-calling tools
guideline_a · guideline_b
call both → structured choice

Captured per case

selected tool · confidence
· free-text rationale

4 What we measure

- Detection accuracy
- Position bias
- Confidence calibration
- Safety-stratified failure rate

500 vignettes · 12 domains · 10 modification types (4 categories) · 21 LLMs · 10,500 evaluations